

**7)Write a Java program to demonstrate synchronization by allowing two threads to access and update a shared bank account safely.**

**PROGRAM:**

```
class BankAccount {  
    private int balance = 1000;  
  
    synchronized void deposit(int amount) {  
        balance += amount;  
        System.out.println(Thread.currentThread().getName()  
            + " deposited: " + amount  
            + " | Balance: " + balance);  
    }  
  
    synchronized void withdraw(int amount) {  
        if (balance >= amount) {  
            balance -= amount;  
            System.out.println(Thread.currentThread().getName()  
                + " withdrew: " + amount  
                + " | Balance: " + balance);  
        } else {  
            System.out.println(Thread.currentThread().getName()  
                + " - Insufficient balance");  
        }  
    }  
}  
  
class Main {  
    public static void main(String[] args) {
```

```
BankAccount account = new BankAccount();
```

```
Thread t1 = new Thread(() -> {  
    account.deposit(500);  
    account.withdraw(200);  
}, "Thread-1");
```

```
Thread t2 = new Thread(() -> {  
    account.withdraw(300);  
    account.deposit(400);  
}, "Thread-2");
```

```
t1.start();  
t2.start();  
}  
}
```

OUTPUT:

Thread-1 deposited: 500 | Balance: 1500

Thread-1 withdrew: 200 | Balance: 1300

Thread-2 withdrew: 300 | Balance: 1000

Thread-2 deposited: 400 | Balance: 1400